

DOM PROJECT

Performance Testing Strategy

Operational Portal / Databricks Pipelines — Structured Data

Scope: Structured CSV Data

Primary Target: 500 GB / 10M Rows / < 4 Hours

Execution: API-triggered Databricks Pipelines

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Introduction

This document defines the performance testing strategy for the DOM structured data ingestion and transformation workflow. The objective is to validate that the platform can process large healthcare datasets efficiently, reliably, and within agreed performance expectations.

The primary focus of this strategy is the backend data processing capability, including API-triggered pipeline execution, Azure Storage input datasets, manifest-based ingestion, Databricks processing, validation logic, quarantine handling, and target data availability in Unity Catalog or downstream tables.

This strategy is not intended to measure general user interface responsiveness. UI interactions may be used only as a supporting mechanism when required. Formal performance validation should be executed through backend APIs and monitored through Databricks, Spark UI, Azure monitoring, application logs, and test execution tools such as JMeter.

Current scope: structured data, primarily CSV-based ingestion. Unstructured data performance testing may be addressed in a separate strategy or future extension once processing rules and acceptance criteria are confirmed.

1. Objectives

Baseline SLA

500 GB

10 million rows processed in less than 4 hours.

2x Scalability

1 TB

20 million rows within the agreed degradation threshold.

Concurrency

2 / 4 / 8+

Validate simultaneous pipeline execution limits.

1.1 SLA Validation

Validate that ingestion and transformation meet the expected business performance requirements. The current baseline target is 10 million rows and 500 GB total dataset size processed in less than 4 hours. The 2x scenario targets 20 million rows and 1 TB total dataset size within the agreed degradation threshold.

Pending confirmation: the current interpretation is that the 2x scenario should not exceed approximately 20% additional processing time compared to the baseline target.

1.2 Scalability Assessment

- Identify maximum validated dataset size.
- Measure performance degradation as volume increases.
- Determine whether processing time scales predictably.
- Identify the volume threshold where SLA compliance is no longer achieved.
- Identify technical limits before job failure, cluster instability, or excessive cost.

1.3 Concurrency Assessment

- Determine whether executions run in parallel, sequentially, or through a hybrid model.
- Measure the maximum sustainable concurrent pipeline executions.
- Assess the impact of concurrency on processing time, Databricks jobs, cluster resources, storage I/O, scheduler behavior, and target table writes.

1.4 Stability, Reliability, Cost Efficiency, Recovery

Validate that large datasets can be processed without critical failures, excessive Spark spill, storage throttling, timeout failures, corrupted target data, or incorrect status updates. The test effort must also measure DBU consumption, estimate cost per run, and validate cleanup and recovery after heavy or failed executions.

2. Scope

2.1 In Scope

- Structured CSV ingestion.
- Manifest-based data ingestion.
- API-triggered pipeline execution.
- Databricks job/pipeline execution.
- Data validation and quality checks.
- Quarantine handling for invalid records.
- Load, stress, and concurrency testing.
- Spark, Databricks, Azure, and application-level monitoring.
- Cleanup and recovery validation.

2.2 Out of Scope

- Full browser/UI performance testing.
- Self Service Portal upload performance testing.
- Selenium or Playwright-based browser performance automation.
- Unstructured data performance testing.
- Production execution without explicit approval.
- Security, penetration, access control, or compliance testing.

3. Key Assumptions

1. Test datasets will be available in Azure Storage before pipeline execution.
2. Performance testing will trigger ingestion through backend APIs or pipeline endpoints.
3. The Operational Portal or backend services expose endpoints to create/configure pipelines, pass manifest or folder locations, trigger Databricks execution, and retrieve execution status.
4. The baseline requirement applies to a full dataset/folder, not necessarily a single file.
5. Test datasets may include one large file, many small files, or mixed file distributions.
6. Databricks is expected to support parallel execution, but the actual concurrency model must be confirmed.
7. The performance environment must be stable and sufficiently representative to produce meaningful results.
8. Test data cleanup is required after execution to control storage costs.
9. DBU consumption and execution metrics must be captured wherever technically possible.

4. Performance Acceptance Criteria

4.1 Baseline Acceptance Criteria

Area	Criteria
Dataset type	Structured CSV data
Dataset size	500 GB total dataset/folder size
Row count	10 million rows
Execution type	Single pipeline execution
Expected result	Pipeline completes successfully
Target time	Less than 4 hours
Data validation	Quality checks complete successfully
Target load	Valid data is loaded to Unity Catalog or target tables
Invalid data	Invalid records are routed to quarantine or reported correctly
Status reporting	Final pipeline status is updated correctly

4.2 Scalability Acceptance Criteria

Area	Criteria
Dataset type	Structured CSV data
Dataset size	1 TB total dataset/folder size
Row count	20 million rows
Execution type	Single pipeline execution
Expected result	Pipeline completes successfully
Target time	To be formally confirmed
Current interpretation	No more than approximately 20% over the baseline processing target
Status	Requires client and stakeholder validation

4.3 Concurrency Acceptance Criteria

Area	Criteria
Dataset size	500 GB per pipeline execution
Concurrent executions	2, 4, 8, or as agreed
Expected behavior	Pipelines execute in parallel, queue, or follow the defined architecture behavior
Stability expectation	No scheduler failure, cluster crash, data corruption, or incomplete status update

Area	Criteria
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Target time	To be confirmed based on agreed concurrency SLA
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5. Test Approach

5.1 Dataset-Based Testing

Performance scenarios should be defined by total dataset or folder size rather than individual file size only. This aligns with the expected pipeline behavior, where the process receives a manifest or folder location and processes all included files.

Baseline Dataset

Folder totaling 500 GB and approximately 10 million rows.

2x Dataset

Folder totaling 1 TB and approximately 20 million rows.

Edge Datasets

One 50 GB CSV, 50,000 small CSV files, and mixed-size datasets.

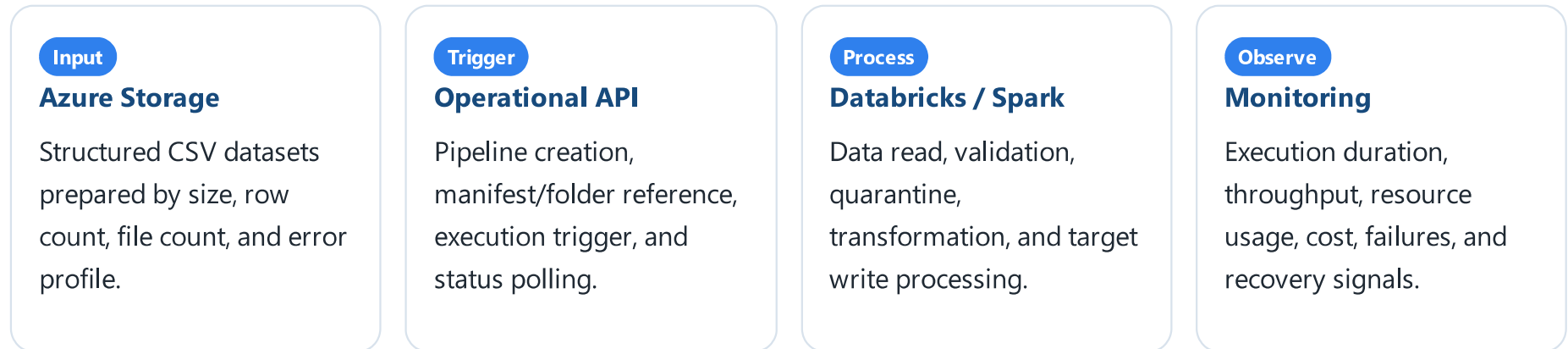
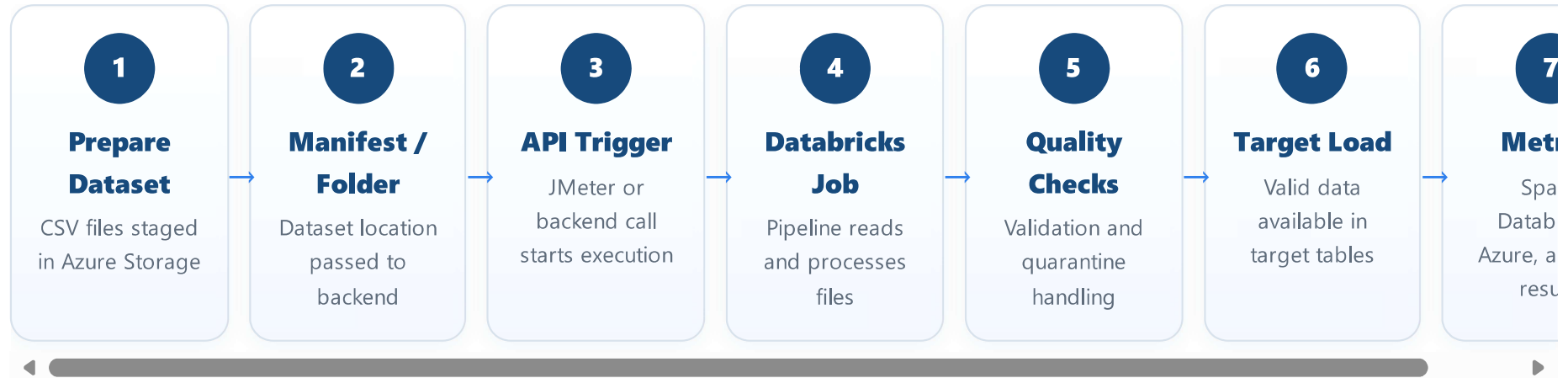
5.2 API-Based Execution

Formal performance testing should use API-based execution whenever possible. JMeter should be used to trigger pipeline creation/configuration, pass manifest or dataset locations, trigger Databricks execution, simulate concurrent execution, and capture API-level response times and failures.

5.3 Backend Monitoring

API response time alone is not sufficient. Backend execution metrics must be captured from Databricks, Spark, Azure, and application logs, including job duration, stage duration, retries, CPU/memory utilization, shuffle, spill to disk, storage throughput, write duration, cluster startup time, DBU consumption, and estimated cost.

6. Execution Flow



7. Test Scenarios

7.1 Scenario Matrix

Scenario	Load Profile	Execution Type	Technical Objective
Baseline	500 GB / 10M rows	Single pipeline	Validate minimum SLA: processing under 4 hours
Scalability 2x	1 TB / 20M rows	Single pipeline	Validate behavior at double volume
Incremental Load	2 TB, 3 TB, 4 TB+	Single pipeline	Identify scaling curve and maximum sustainable volume
Large File	Dataset includes one 50 GB CSV file	Single pipeline	Validate large individual file handling
Small Files	50,000 small CSV files	Single pipeline	Measure file fragmentation and metadata overhead
Mixed Dataset	Small, medium, and large CSV files	Single pipeline	Simulate realistic customer delivery patterns
Data Quality	Controlled invalid records	Single pipeline	Measure validation and quarantine impact
Concurrency Baseline	2 concurrent 500 GB executions	Concurrent triggers	Validate behavior under simultaneous execution
Concurrency Growth	4, 8, or more concurrent executions	Concurrent triggers	Identify concurrency limits

Scenario	Load Profile	Execution Type	Technical Objective
Stress	Increasing volume and/or concurrency	Single and concurrent	Identify failure threshold and degradation point
Recovery	Re-run after heavy or failed execution	Post-stress validation	Validate system recovery
Cleanup	Remove test artifacts	Post-test execution	Validate repeatable cleanup and cost control

7.2 Scenario Details

Baseline Structured Data Test
Input: 500 GB structured CSV, approximately 10M rows.
Success: completes under 4 hours, loads target data, handles invalid records, and updates final status.

Scalability Test — 2x Volume
Input: 1 TB structured CSV, approximately 20M rows.
Success: completes within agreed threshold without instability or excessive retries.

Incremental Volume Test
Run 500 GB, 1 TB, 2 TB, 3 TB, and higher volumes if approved to identify SLA threshold and production load recommendations.

Large File Edge Case
Validate processing of a dataset containing at least one 50 GB CSV without memory overflow or unacceptable Spark spill.

Small File Fragmentation

Data Quality and Quarantine

Evaluate behavior with approximately 50,000 small CSV files and measure enumeration, driver, metadata, and I/O overhead.

Measure validation overhead using 100% clean data and datasets with 1%, 5%, and 10% invalid records.

Concurrency Test

Trigger 2, 4, and 8 simultaneous executions, if approved, and capture start time, queue time, execution time, completion time, and final status.

Stress, Recovery, and Cleanup

Increase load until SLA breach or failure, validate recovery behavior, and confirm all test artifacts are cleaned up.

8. Positive and Negative Test Coverage

8.1 Positive Cases

- Process clean structured CSV data.
- Process 500 GB dataset within SLA.
- Process 1 TB dataset within agreed scalability threshold.
- Process datasets with mixed file sizes.
- Process high file-count datasets.
- Execute multiple pipelines simultaneously when supported by architecture.

8.2 Negative and Edge Cases

- Corrupted or malformed records.
- Missing headers and incorrect delimiters.
- Invalid date formats and duplicate records.
- High null percentage.
- One very large file and 50,000 small files.
- Concurrent execution beyond supported limits.
- Resource-constrained execution, if approved.
- Pipeline interruption followed by recovery validation.

9. Metrics

Category	Metrics
End-to-End	Total trigger-to-completion time, input-to-target availability, API-to-job start time, queue time, execution time per pipeline, execution time per concurrent run.
Throughput	Records processed per second, GB processed per hour, files processed per minute, valid records loaded per second, invalid records quarantined per second.
Data Quality	Total records, valid records, invalid records, invalid percentage, quarantine processing time, quality check overhead.
Databricks / Spark	Job duration, stage duration, task duration, CPU, memory, shuffle, spill to disk, failed tasks, retries, cluster startup time, autoscaling, worker utilization.
Storage and Target	Azure Storage read throughput, throttling, target write duration, target table creation/update time, metadata operation time, file enumeration time.
Cost	DBU consumption per run, estimated cost per baseline, estimated cost per 1 TB run, cost per concurrent group, cost by cluster configuration, cost impact of failures or retries.

10. Tooling Strategy

JMeter

Recommended for formal API-based performance execution, concurrent triggering, repeatable test plans, response time capture, error rate measurement, and execution reports.

Postman

Recommended for initial endpoint validation, authentication checks, payload confirmation, and small-scale exploratory execution.

Databricks and Spark UI

Used to analyze execution behavior, bottlenecks, resource utilization, stage delays, memory pressure, shuffle/spill behavior, failures, and retries.

Azure Monitoring

Used to track storage read/write activity, network throughput, infrastructure health, resource throttling, and cost indicators where available.

11. Execution Plan

Phase	Name	Key Activities
0	Requirements and Environment Readiness	Confirm SLA interpretation, 2x expectation, concurrency criterion, execution model, endpoints, authentication, environment stability, cluster configuration, monitoring access, and cleanup process.
1	Dataset Preparation	Prepare 500 GB, 1 TB, 50 GB file, 50,000 small files, mixed-size, invalid-record, and optional larger datasets. Document folder path, manifest, size, file count, row count, schema, and error profile.
2	JMeter Test Plan Preparation	Create suites for single execution, baseline, 2x, concurrency, status polling, and optional cleanup. Parameterize environment URL, auth token, customer/project IDs, manifest, job ID, thread count, and iterations.
3	Baseline Execution	Run the 500 GB / 10M rows scenario, capture API, Databricks, Spark, storage, and cost metrics, and compare against SLA.
4	Scalability Execution	Run the 1 TB / 20M rows scenario, compare against agreed 2x criteria, and document scaling behavior.
5	Edge Case Execution	Run 50 GB individual file, 50,000 small files, mixed dataset, and data quality overhead tests.
6	Concurrency Execution	Run 2, 4, and 8 concurrent pipelines if approved. For pure concurrency, start threads simultaneously or as close as technically possible.
7	Stress and Breaking Point	Increase load until SLA breach, pipeline failure, cluster instability, excessive cost, storage throttling, or unacceptable retry/failure rate.

Phase	Name	Key Activities
8	Recovery and Cleanup	Validate system health, execute subsequent pipeline validation, clean test data, remove temporary files, isolate/remove target test records, and confirm cleanup completion.

12. Risks and Mitigation

Risk	Impact	Mitigation
Performance environment is unstable or unavailable	Delays execution and may invalidate results	Confirm environment readiness before formal testing
Baseline SLA interpretation is ambiguous	Incorrect pass/fail decisions	Obtain formal stakeholder confirmation
2x + 20% criterion is unclear	Misaligned expectations	Confirm whether the threshold applies to total time, throughput, or degradation
Pipeline concurrency model is unknown	Incorrect concurrency test design	Confirm whether execution is parallel, queued, or hybrid
High DBU consumption	Increased cost	Use approved job clusters, define test windows, and limit unnecessary iterations
Large datasets increase storage cost	Cost overrun	Implement cleanup process and storage retention rules
Self Service Portal is not ready	Upload performance cannot be tested	Limit current scope to backend pipeline triggering
Test data generation is delayed	Blocks execution	Assign dataset ownership early
Non-production environment differs from production	Results may not represent production behavior	Document environment differences and avoid overgeneralization

Risk	Impact	Mitigation
Other processes run during testing	Noisy or unreliable results	Coordinate clean execution windows
Missing monitoring access	Incomplete root cause analysis	Confirm access to Databricks, Spark UI, Azure logs, and app logs before execution

13. Open Questions

Business and Acceptance Criteria

1. Does the 500 GB / 10M rows / < 4 hours SLA apply to one pipeline execution only?
2. Does the 2x scenario mean exactly 1 TB / 20M rows?
3. Does the 20% threshold mean the 1 TB scenario must finish within approximately 4 hours and 48 minutes?
4. Is concurrency part of the formal acceptance criteria?
5. How many concurrent clients or projects should be expected in production?
6. What is the maximum expected dataset size per customer delivery?
7. Should structured and unstructured data follow the same SLA?

Technical and Architecture

8. Are Databricks pipelines executed in parallel, queued, or through a hybrid model?
9. What is the maximum number of concurrent Databricks jobs allowed?
10. Are there cluster, workspace, or job-level concurrency limits?
11. Are there API rate limits or throttling rules?
12. What timeout values exist at API, backend, Databricks, and storage layers?
13. What happens when a pipeline fails?
14. Is retry behavior automatic or manual?
15. Are cleanup, delete, or restore endpoints available?

Test Data

16. Who will generate the 500 GB and 1 TB datasets?
17. Will test data be synthetic, anonymized production-like data, or real de-identified data?
18. What schema should the CSV test files use?

Observability

21. Who has access to Databricks and Spark UI during test execution?
22. Where are pipeline execution logs stored?

19. What percentage of invalid records should be included in negative tests?

20. Should datasets include healthcare-like dummy data or generic test data?

23. How is final pipeline status updated in the Operational Portal?

24. Can DBU consumption be captured per run?

25. Can storage throughput and throttling be monitored during execution?

14. Deliverables

Deliverable	Description
Performance Test Plan	Complete plan covering scope, assumptions, scenarios, criteria, dataset definitions, execution approach, tools, risks, and open questions.
JMeter Test Suite	Reusable assets for pipeline configuration, execution, status polling, concurrent execution, and optional cleanup calls.
Dataset Inventory	Dataset documentation including folder path, manifest path, total size, file count, row count, schema, error profile, and purpose.
Results Matrix	Technical results by scenario, including dataset size, execution time, throughput, DBU usage, cost estimate, cluster configuration, pass/fail status, and bottlenecks.
Executive Performance Report	Stakeholder-facing summary of SLA compliance, maximum validated volume, concurrency capacity, risks, bottlenecks, production limits, and cluster/job recommendations.
Recommended Configuration Guide	Final technical recommendations for cluster size, autoscaling, job cluster usage, maximum dataset size, maximum concurrent executions, cleanup strategy, and monitoring requirements.

15. Recommended Execution Priority

- 1 Confirm baseline SLA interpretation.
- 2 Confirm 2x scalability expectation.
- 3 Confirm API endpoints and execution model.
- 4 Prepare the 500 GB structured dataset.
- 5 Execute the baseline performance test.
- 6 Prepare the 1 TB structured dataset.
- 7 Execute the 2x scalability test.
- 8 Execute high-risk edge cases: 50 GB individual file and 50,000 small files.
- 9 Execute initial concurrency tests with 2 and 4 simultaneous executions.
- 10 Capture cost and Databricks metrics.

- 11 Produce the initial performance findings report.

Executive Summary

This strategy validates the structured data ingestion and transformation capabilities of the DOM platform using dataset-based performance testing. The primary baseline target is to process **500 GB / 10 million rows in less than 4 hours**, followed by a **1 TB / 20 million row scalability test** and additional stress, edge case, concurrency, recovery, and cleanup validations.

The testing approach prioritizes backend API-triggered pipeline execution, Databricks/Spark observability, cost measurement, and production-readiness recommendations. The final outcome will define validated processing limits, scalability behavior, concurrency capacity, cost implications, and recommended operating parameters for production.

Performance Testing Strategy — DOM Project — Operational Portal / Databricks Pipelines — Structured Data